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Recognition of Manhole Cover Status for Refuse Collection Vehicles Utilizing YOLO26-ABF

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Abstract

Refuse collection vehicles are essential for urban sanitation. Adverse conditions, including damaged, displaced, or absent maintenance hole covers on roadways, can result in safety incidents—such as equipment jamming or overturning—during operations, consequently impacting the continuity and safety of cleaning activities. To facilitate real-time, precise identification of maintenance hole cover conditions during cleaning operations, this study proposes a detection method using an enhanced YOLO26 model. Through the development of the YOLO26-ABF detection model, which integrates multiscale feature fusion and multidimensional data augmentation techniques, we attained precise identification of five maintenance hole cover conditions—"good," "broken," "circle," "loose," and "uncovered"—resulting in a moderate rise in parameter count and computational demand. The experimental findings indicate that this method achieves substantial improvements over the baseline model on a custom dataset. It can precisely identify the condition of maintenance hole covers, providing reliable status awareness for waste collection vehicles and ensuring the safe and stable execution of cleaning activities.

Keywords: refuse collection vehicles; maintenance hole cover condition; YOLO26-ABF; multiscale feature fusion; multidimensional data augmentation techniques

1. Introduction

The integrity of urban road infrastructure is essential for maintaining municipal management quality and public safety. Maintenance hole covers, functioning as ground-level entry points to urban drainage, electrical, and communication systems, are extensively located throughout city streets and alleys [1]. The rapid advancement of smart cities has rendered the real-time monitoring and intelligent control of maintenance hole cover status essential for improving urban governance [2,3]. In urban sanitation operations, garbage collection trucks are becoming increasingly vital [4–6]. However, if maintenance hole covers on the roadway are broken, misaligned, loose, or uncovered, they can easily lead to severe safety incidents during cleaning operations, including equipment jamming, chassis damage, or even vehicle rollovers. These accidents impede operational continuity and jeopardize the safety of adjacent people and traffic [7]. Consequently, developing a detection algorithm for integration into the visual systems of street sweeper vehicles to enable real-time, precise assessment of maintenance hole cover quality has considerable academic and engineering significance.

Check for updates

Academic Editor: Atsushi Mase

Received: 17 July 2026
Revised: 22 August 2026
Accepted: 23 August 2026
Published: 26 August 2026

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Initial investigations into maintenance hole cover surveillance predominantly used Internet of Things (IoT) technology, employing sensor modules based on NB-IoT or LoRa to assess the tilt and displacement of maintenance hole covers [8]. While these methods can yield high-frequency status notifications, they are expensive to implement, challenging to sustain, and fail to intuitively capture the specifics of damage on the maintenance hole cover surface. Consequently, researchers have sought to use mobile laser scanning (MLS) data or LiDAR systems to automate the identification of road maintenance hole covers [9,10]. While these methods offer benefits for three-dimensional spatial localization, their significant data processing complexity and the cost of the required equipment hinder large-scale implementation in sanitation vehicle fleets. Conversely, computer vision-based detection methods have increasingly emerged as the predominant approach in this domain owing to their affordability, efficiency, and comprehensive information content [11].

In recent years, deep learning algorithms, exemplified by the YOLO series, have attained notable success in road surface object recognition [12,13]. Regarding maintenance hole cover detection, existing studies have achieved good classification accuracy on specific datasets by modifying the YOLOv5, YOLOX, or the latest YOLOv8 frameworks [14–17]. For example, Ding et al. used an improved YOLOv8 to enhance the efficiency of urban maintenance hole cover defect management [18]; Zeng et al. proposed MCPH-YOLO, which was optimized for detecting hidden defects in complex scenes [19]. Pang et al. proposed a real-time detection model based on an improved YOLO, which effectively enhanced the recognition efficiency of maintenance hole covers under conventional viewpoints by integrating lightweight modules with attention mechanisms [20]; Mattheuwsen et al. explored the use of transfer learning for recognition on mobile mapping point cloud data, alleviating the issue of sample scarcity under specific viewpoints [21]. However, in the operational environment of garbage collection trucks, maintenance hole cover detection still faces two major challenges: first, it is difficult to detect small maintenance hole covers. Camera mounting angles and operational speeds result in maintenance hole covers at a distance occupying minimal pixels in the image, leading to semantic information loss in traditional feature pyramids (PAN) during multiscale fusion [22]. Furthermore, there exists a significant data imbalance; acquiring real-world image samples of highly hazardous conditions—such as damaged or absent maintenance hole covers—is challenging, thereby constraining the model's generalization capability for these crucial classes [23].

Among these studies, MCPH-YOLO [19] and several improved YOLOv8/YOLOX detectors [12,15,18] have demonstrated the effectiveness of attention, small-object handling, and multi-scale feature fusion for manhole cover detection. However, these methods were developed for general road or defect inspection settings and do not target the five-state condition taxonomy from the onboard viewpoint of refuse collection vehicles. They also do not combine the specific BiFPN + P2 + ASFF architecture with a synthetic-image and pseudo-labeling pipeline under an extreme class-imbalance setting. The contribution of this work therefore lies in the task formulation and the integrated solution, rather than in any single architectural component.

This study proposes a method for assessing maintenance hole cover conditions utilizing an enhanced YOLO26 and develops the YOLO26-ABF detection model. Initially, at the architectural level, the traditional PAN structure is substituted with a Bidirectional Feature Pyramid Network (BiFPN), which improves weight distribution in feature extraction via weighted bidirectional connections. A higher-resolution P2 feature layer is introduced to extract low-level spatial characteristics specifically for small maintenance hole cover targets. Additionally, an Adaptive Spatial Feature Fusion (ASFF) mechanism is implemented to efficiently eliminate conflicting information during fusion, thereby substantially enhancing the detection accuracy of small targets. Furthermore, to tackle the challenge of

sample scarcity, this paper utilizes augmentation techniques such as Mosaic, MixUp, and Copy–Paste, while also generating synthetic images to replicate rare scenarios—such as absent or misaligned maintenance hole covers—and integrates these with semi-supervised learning (pseudo-labeling) to broaden the training set, consequently improving the model's robustness in complex lighting and occlusion conditions.

This paper's primary contributions are as follows:

(1) In the context of garbage truck operations, we developed a high-quality dataset of maintenance hole covers categorized into five states: "good," "broken," "circle," "loose," and "uncovered." Synthetic images were generated to correct the severe imbalance of the rare hazardous states, and semi-supervised pseudo-labeling was additionally used to enrich the diversity of scenarios and road-surface materials.

(2) We propose YOLO26-ABF for five-state manhole cover condition recognition from the onboard viewpoint of refuse collection vehicles. The method adaptively combines BiFPN, a P2 high-resolution layer, and ASFF to address small-object detection while incurring a tolerable increase in computational demand; this combination, rather than any single pre-existing component, is tailored to the target operational setting.

(3) The experimental results indicate that the enhanced algorithm achieves an optimal balance between detection accuracy and computational efficiency. End-to-end latency measurements on an RTX 5060 Ti show that YOLO26-ABF processes frames at 51 FPS on average, with a p50 latency of 13 ms and a p95 latency of 44 ms. Because the p95 latency exceeds the 33.3 ms frame interval of the 30 FPS camera, the real-time claim is based on the average latency rather than the worst-case tail.

2. Materials and Methods

2.1. Comprehensive Design of the Detection Framework

This study proposes a visual detection framework that enables garbage collection trucks to achieve real-time, precise identification of maintenance hole cover anomalies in intricate sanitation operations. The framework incorporates multidimensional data augmentation and multiscale feature fusion, following a systematic approach of "data-driven enhancement—detailed feature extraction—real-time status recognition." It aims to resolve industry challenges, including low recognition rates for small maintenance hole covers and insufficient samples for hazardous operating conditions in the garbage truck's environment. The detection framework established in this study is illustrated in Figure 1, with the independently created YOLO26-ABF detection model as its basic component.

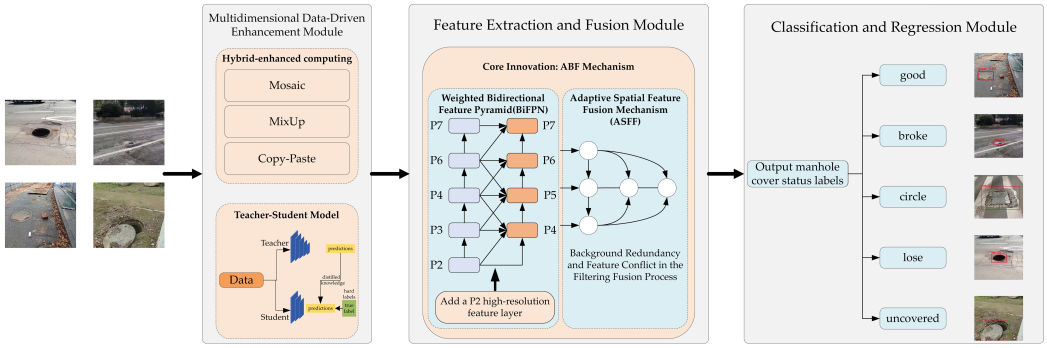


Figure 1. Comprehensive framework for YOLO26-ABF manhole cover condition assessment.

The complete perception system fundamentally comprises three functional modules that operate in unison.

Multidimensional Data-Driven Module (Scheme D): To tackle the challenge of extreme sample distribution imbalance in maintenance hole cover detection, a tripartite approach

of "basic augmentation + synthetic simulation + semi-supervised expansion" was devised. This module employs Mosaic, MixUp, and Copy–Paste techniques to enhance background complexity. Synthetic image technology is used to recreate rare incident scenarios such as "loose" and "uncovered," thereby directly correcting the imbalance of high-risk samples, while the pseudo-labeling method from semi-supervised learning is additionally incorporated to exploit unlabeled road-surface data, expanding the diversity of scenarios and road-surface materials and further enhancing the model's recognition robustness under complex lighting and dynamic conditions.

The Feature Extraction and Fusion Module (Backbone & Neck, Scheme B) uses a lightweight YOLO26 backbone network to extract initial road surface information. Real-time performance was measured on an RTX 5060 Ti, while deployment on embedded vehicle hardware has not yet been measured. Subsequently, a weighted bidirectional feature pyramid (BiFPN) reconstructs the feature flow path, and a P2 high-resolution feature layer is incorporated to capture the intricate spatial details of small maintenance hole covers. The integrated adaptive spatial feature fusion (ASFF) mechanism concurrently eliminates background redundancy and feature conflicts during fusion, enabling the model to precisely identify targets amid intricate background noise, such as oil stains on the road and fallen leaves.

The Status Classification and Regression Module (Head) generates five-class status labels for maintenance hole covers, along with their positional coordinates. The recognition outcomes serve as the foundational input for the autonomous obstacle-avoidance control system of the garbage truck, providing reliable perceptual support for safe, seamless operation.

2.2. Enhancement of the YOLO26-ABF Network Architecture

This study designates the nano-scale YOLO26 model (YOLO26-n) as the detection benchmark [24,25]. YOLO26 is a single-stage object detector designed for edge-embedded environments, featuring NMS-free end-to-end inference through a one-to-one detection head and omitting the distribution focal loss. In this study, the backbone and feature extraction follow the official YOLO26 implementation, while the original neck is replaced by the proposed BiFPN and a P2 high-resolution layer is added. The three detection heads are constructed on the P3, P4, and P5 levels. Because the one-to-one head assigns at most one prediction to each object, the final outputs are decoded without non-maximum suppression. The ASFF modules inserted before each head operate at the feature level and therefore do not alter the one-to-one assignment or the NMS-free decoding procedure.

The YOLO26-ABF detection model created in this study incorporates a high-resolution P2 feature layer to enhance the low-level edge texture information that is diminished during deep downsampling, specifically targeting the feature loss of small maintenance hole covers at extended distances; however, the addition of the P2 layer exacerbates the disparities in receptive fields among features at varying levels. Direct fusion may lead to semantic discrepancies and information dilution; hence, a BiFPN-based weighted bidirectional feature pyramid was implemented to replace the original PAN structure. The learnable weight allocation mechanism optimized the aggregation paths of multiscale information flows, improving feature fusion efficiency and effectiveness. Ultimately, the ASFF (Adaptive Spatial Feature Fusion) mechanism is incorporated before each detection head to dynamically suppress background noise and enhance target features in pixel space, thereby mitigating semantic conflicts during multiscale fusion. Collectively, these three elements provide a synergistic optimization system that addresses the perceptual challenges inherent in the onboard viewpoint of street sweepers, as shown in Figure 2, achieving an optimal balance between detection precision and computational cost.

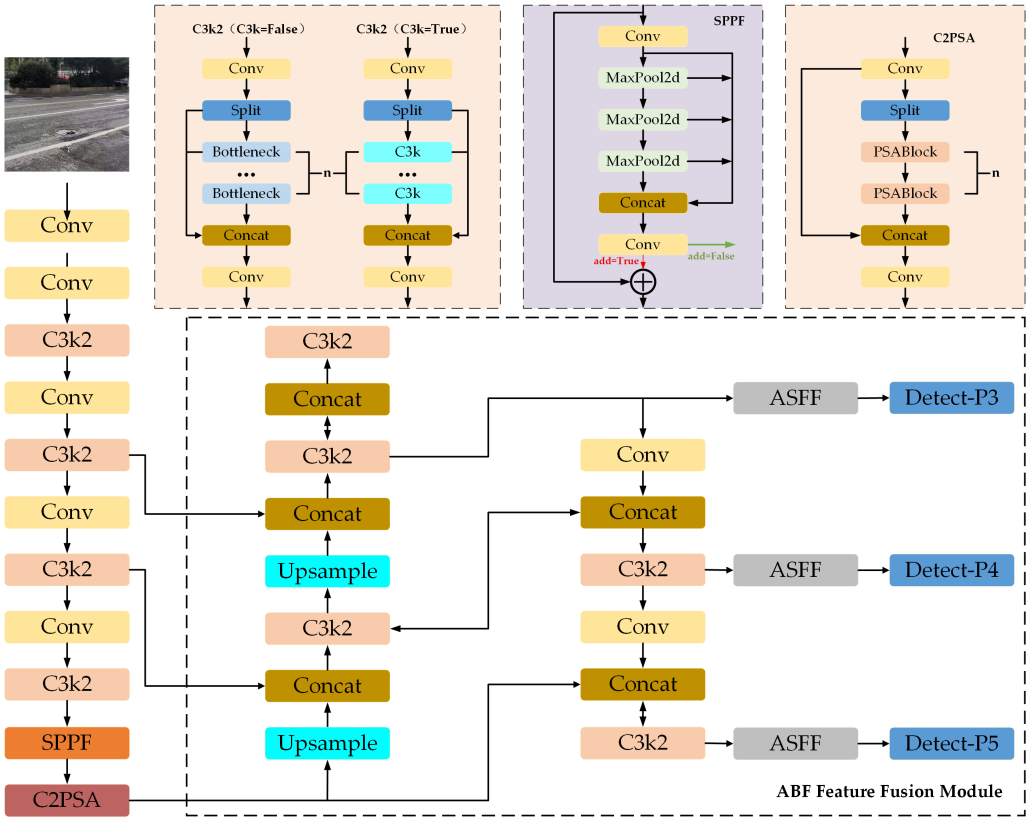


Figure 2. Schematic representation of the YOLO26-ABF network architecture.

2.2.1. Weighted Bidirectional Feature Pyramid Network (BiFPN)

Conventional feature pyramids (PANs) generally use basic cascades or uniformly weighted summation to process multiscale information flows, thereby overlooking the varied contributions of features at distinct levels to the ultimate detection outcomes [26]. This study presents the Weighted Bidirectional Feature Pyramid Network (BiFPN) to enhance feature aggregation efficiency.

BiFPN facilitates bidirectional feature flow between layers through top-down and bottom-up pathways and incorporates a weight allocation technique grounded in Fast Normalized Fusion [27]. The mathematical formulation of the fusion process for any feature node at a specific layer is as follows:

$$O = \sum_i \frac{w_i}{\epsilon + \sum_j w_j} \cdot I_i \tag{1}$$

where I_i denotes the input feature map, O represents the fused output feature, and w_i signifies the weight parameter autonomously acquired by the model during training. To maintain non-negative weights, the model employs the ReLU activation function before the weights, ensuring that $w_i \geq 0$. The parameter ϵ is established at 0.0001 to guarantee numerical stability. This mathematical approach enables BiFPN to automatically enhance the weights of high-contribution feature layers, thereby effectively addressing the feature dilution issue that occurs when integrating shallow-layer spatial information with deep-layer semantic information.

2.2.2. P2 High-Resolution Feature Layer for the Detection of Minuscule Objects

When a garbage truck is in motion, maintenance hole covers at a distance often appear like small targets. In the conventional YOLO architecture, following several iterations of downsampling, the low-level geometric details of extremely small targets are nearly entirely

diminished. Zeng et al. highlighted in their research on small-target detection in drone imagery that incorporating shallow feature maps with high spatial resolution, such as the P2 layer, can effectively capture the nuanced geometric details of targets, thereby mitigating the challenges of feature dilution and loss during deep-layer downsampling [28]. Consequently, this study integrated a P2 high-resolution feature layer into the feature extraction stage. Through the upsampling of deep semantic features and their fusion with shallow features across various scales, the P2 layer effectively retains high-frequency texture information—such as the circular edges and damage cracks of maintenance hole covers—thereby significantly improving the model's sensitivity to nuanced visual distinctions among the "good," "broken," and "circle" states.

2.2.3. Mechanism of Adaptive Spatial Feature Fusion (ASFF)

Due to the varying receptive fields of layers P2–P5, direct fusion may result in semantic conflicts, specifically interference between redundant background information and target features [29]. The network generates four feature levels (P2–P5), while the three detection heads are constructed on the P3, P4, and P5 levels. For every head, ASFF is applied before the prediction head and fuses the three aligned inputs, which are the P3, P4, and P5 features; the P2 level is an additional high-resolution feature used by BiFPN to enrich the P3-level features and is not an independent ASFF input. Let the multiscale features transmitted from the Neck to the prediction layer be denoted as x^1, x^2, x^3; following scale transformation and alignment, the resultant features are represented as $x^{n \to l}$. The feature fusion aggregation formula for ASFF at spatial point (i, j) in layer l is articulated as

$$y_{ij}^l = \alpha_{ij}^l \cdot x_{ij}^{1 \to l} + \beta_{ij}^l \cdot x_{ij}^{2 \to l} + \gamma_{ij}^l \cdot x_{ij}^{3 \to l} \tag{2}$$

Here, y_{ij}^l denotes the output feature subsequent to adaptive fusion; α_{ij}^l, β_{ij}^l, and γ_{ij}^l represent spatial weights independently acquired by the network. To guarantee that the aggregate of the weights across all scales equals 1, ASFF employs the Softmax function to regulate the computation:

$$\alpha_{ij}^l = \frac{e^{\lambda_{\alpha_{ij}}^l}}{e^{\lambda_{\alpha_{ij}}^l} + e^{\lambda_{\beta_{ij}}^l} + e^{\lambda_{\gamma_{ij}}^l}} \tag{3}$$

Here, $\lambda_\alpha, \lambda_\beta, \lambda_\gamma$ represent control parameters generated by the 1×1 convolutional layer. This adaptive filtering mechanism allows the model to dynamically attenuate background noise—such as fallen leaves and oil stains on the road surface—by assigning them minimal weights, while simultaneously reallocating weights to the high-resolution layers upon detecting small maintenance hole covers, thereby markedly improving its discrimination capability in intricate backgrounds.

2.3. Strategy for Enhancement and Robustness Improvement via Multidimensional Data Analysis

This work presents a data reconstruction technique that integrates "multidimensional augmentation computation" with "semi-supervised learning" to address the training imbalance caused by the severe paucity of samples depicting maintenance hole cover anomalies (e.g., "loose" or "uncovered").

2.3.1. Hybrid Data Augmentation and Synthetic Image Generation

This research incorporates Mosaic, MixUp, and Copy–Paste data augmentation methods in the input phase. Using MixUp as an illustration, this approach broadens the feature distribution space by pixel-level linearly interpolating two randomly chosen images and their corresponding labels [30]. The fundamental computational logic is as follows:

$$\tilde{x} = \lambda x_m + (1 - \lambda) x_n \tag{4}$$

$$\tilde{y} = \lambda y_m + (1 - \lambda) y_n \tag{5}$$

Let x_m and x_n denote two randomly picked input image matrices, while y_m and y_n represent their associated one-hot true labels. Additionally, λ is distributed according to a Beta distribution, specifically $\lambda \sim Beta(\alpha, \alpha)$, where α is a hyperparameter.

Synthetic images were generated by compositing real maintenance hole cover templates onto road-surface backgrounds. A total of 42 cover templates (22 for "loose" and 20 for "uncovered") and 75 background images were used. Each template was transformed by perspective warping with a rotation range of ± 15 degrees, a scale range of 0.8–1.2, and a translation of up to ± 50 pixels, and was then blended into the background using Poisson image editing. Templates were placed only within the road area, with a minimum distance from the image border and without overlapping other covers. All generated images were manually inspected; images with visible artifacts, unrealistic illumination, or inconsistent geometry were discarded. After quality control, 260 "loose" and 255 "uncovered" synthetic images remained. This approach directly addresses the significant challenges of collecting "loose" and "uncovered" samples in real-world contexts and refines the decision boundaries across multiple classes, thereby enhancing the model's resilience to overlapping objects and illumination variations.

2.3.2. Semi-Supervised Learning Utilizing Pseudo-Label Constraints

This research presents a semi-supervised learning methodology utilizing a teacher–student architecture to maximize the potential of extensive unlabeled road video data. A teacher model, i.e., the best-performing YOLO26 baseline trained on the annotated training set, generates pseudo-labels for the unlabeled frames. The student model is then trained on the combination of labeled and pseudo-labeled data, and the overall loss function L_{total} comprises the supervised loss L_{sup} on labeled samples and the pseudo-label loss L_{unsup} on pseudo-labeled samples [31]:

$$L_{total} = L_{sup}(\theta) + \eta \cdot L_{unsup}(\theta) \tag{6}$$

where θ represents the network parameters and η is the balancing coefficient of the pseudo-label loss. Only pseudo-labels with a teacher prediction confidence p_i above a threshold τ (e.g., $\tau = 0.85$) are retained:

$$L_{unsup} = \frac{1}{N_u} \sum_{i=1}^{N_u} \mathbb{I}(p_i \geq \tau) \left[\mathcal{L}_{cls}(y_{pred,i}, \hat{y}_{cls,i}) + \lambda_{box} \mathcal{L}_{box}(b_{pred,i}, \hat{b}_{box,i}) \right] \tag{7}$$

where $\mathbb{I}(\cdot)$ is the indicator function, $\mathcal{L}_{cls}$ is the classification loss, $\mathcal{L}_{box}$ is the bounding-box regression loss (CIoU), and λ_{box} is the regression weight. The balancing coefficient η was set to 1.0, and the confidence threshold τ was 0.85. In this way, the pseudo-labels generated by the teacher are used as additional supervision for the student model without requiring extra manual annotation. This technique effectively leverages semantic priors from unlabeled data, improving the model's adaptability to the diverse road-surface materials encountered in the collected routes. The 12 h of unlabeled road-surface recordings used for pseudo-labeling were collected on different routes and time periods from those of the training, validation, and test sets, ensuring that no test scene contributed to the pseudo-labeled samples.

This study addresses the challenges of maintenance hole cover recognition in the operational context of garbage collection trucks by developing the YOLO26-ABF visual

detection framework. At the architectural level, the model incorporates a profound fusion mechanism that includes BiFPN, a high-resolution P2 layer, and ASFF, adeptly addressing the problems of feature dilution and semantic conflict for small targets; at the data level, it integrates hybrid augmentation, synthetic images, and semi-supervised pseudo-labeling techniques to mitigate the challenges posed by long-tail distribution resulting from extreme sample imbalance in abnormal operating conditions. The close integration of the previously indicated algorithmic optimizations and data-driven techniques establishes a robust model framework for future experimental validation and performance assessment.

3. Results

3.1. Experimental Dataset and Preprocessing Procedures

This study developed a dataset on the condition of road maintenance hole covers specifically designed for the operating contexts of waste collection vehicles. The raw samples were predominantly gathered in the field by affixing a camera to the front of a garbage collection vehicle, positioned to closely align with the driver's viewpoint, as illustrated in Figure 3, encompassing a range of intricate road-surface backgrounds, including urban main roads, secondary roads, and alleys. The dataset consists of frames extracted from vehicle-mounted recordings collected along multiple operating routes and photographs captured manually on the road. To avoid temporal and spatial leakage between nearly identical consecutive frames, the training, validation, and test sets were constructed at the level of recording routes, and manually captured photographs were grouped by capture location and time; images originating from the same route or location were assigned to a single subset only. All images were acquired by a single vehicle-mounted camera during one collection campaign. Accordingly, the evaluation in this study reflects within-campaign generalization and does not yet cover cross-vehicle, cross-camera, or cross-region conditions.

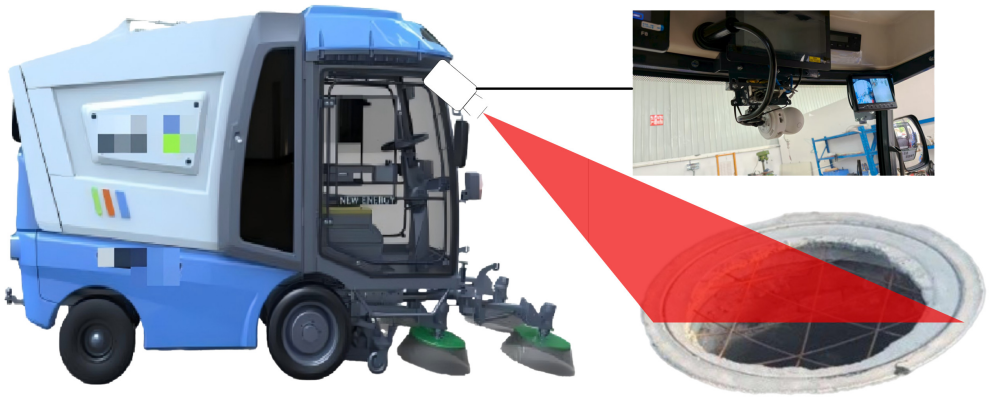


Figure 3. Illustration of intricate road surface maintenance hole cover data acquisition from an onboard viewpoint.

To mitigate the long-tail distribution issue arising from the paucity of samples under abnormal operating conditions, the training set was augmented using synthetic image synthesis and semi-supervised pseudo-labeling. All samples in the validation and test sets are original images captured in the field. Synthetic and pseudo-labeled samples were used for training only and never appeared in the validation or test sets.

3.1.1. Distribution of Classes and Effects of Data Augmentation

The dataset comprises five categories of maintenance hole cover conditions—"good," "broken," "circle," "loose," and "uncovered"—which effectively encompass both standard and atypical scenarios that may arise during sanitation activities, as shown in Figure 4. The

original dataset was partitioned for experimentation into training, validation, and test sets in an 8:1:1 ratio.

Figure 4. Illustrations of samples representing the five goal states of maintenance hole covers: (**a**) good; (**b**) broken; (**c**) circle; (**d**) loose; (**e**) uncovered.

A total of 1673 photos were collected from the original field data. Table 1 illustrates the distribution of sample counts among groups. Intact maintenance hole cover samples constitute the predominant fraction, but high-risk aberrant samples, such as "loose" and "uncovered," exhibit a characteristic long-tailed distribution owing to their infrequent occurrence in real-world contexts.

Table 1 reports the complete sample accounting before and after each augmentation stage. The training set originally contained 1338 images, including 241 high-risk samples ("loose" and "uncovered"), accounting for 18.0% of the training images. To alleviate the severe imbalance of these rare hazardous states, 515 synthetic images (260 "loose" and 255 "uncovered") were generated using spatial perspective transformation and Poisson blending and were added to the training set, increasing the high-risk share to 40.8% (756/1853). In the semi-supervised stage, 920 pseudo-labeled frames were collected from 12 h of unlabeled road-surface recordings and added to the training set to expand the diversity of scenarios and pavement materials. Among them, 860 (93.5%) belong to the three common states and 60 (6.5%) to the high-risk states (31 "loose" and 29 "uncovered"), reflecting the natural distribution of the unlabeled recordings while ensuring that rare hazardous states remained represented. Consequently, the high-risk share of the final

training set is 29.4% (816/2773). Overall, the synthetic stage directly corrected the class imbalance, whereas the pseudo-labeling stage primarily enriched scenario and material diversity and provided semi-supervised regularization.

Table 1. Sample accounting before and after data augmentation by category.

Category	Original Train	Original Val	Original Test	+Synthetic	+Pseudo-Labeled	Final Train	Original Total
good	696	87	87	0	420	1116	870
broken	214	27	27	0	220	434	268
circle	187	23	24	0	220	407	234
loose	120	15	15	260	31	411	150
uncovered	121	15	15	255	29	405	151
Total	1338	167	168	515	920	2773	1673

3.1.2. Experimental Setting and Parameter Configuration

This dataset employs the open source annotation tool LabelMe to execute semantic annotation on every image. The annotation boxes are precisely aligned along the outer margins of the maintenance hole covers, completely encompassing the main body of the cover. The criteria for classification are as follows:

The maintenance hole cover is undamaged, correctly aligned, and securely positioned over the maintenance hole entrance.

The maintenance hole cover surface displays structural degradation, including cracks, chips, or fractures; however, its general location remains unchanged.

The perimeter of the maintenance hole cover is misaligned with the maintenance hole aperture, although the cover adequately conceals the opening.

The maintenance hole cover is entirely dislodged from the maintenance hole aperture, rendering the opening unprotected.

The maintenance hole cover has been displaced or overturned adjacent to the maintenance hole entrance, leaving it inadequately protected.

All samples were annotated manually in accordance with the COCO annotation rules. Figure 5 presents instances of annotated images.

(**a**)

(**b**)

Figure 5. Illustration of annotated images from the road manhole cover dataset: (**a**) original image from the dataset; (**b**) corresponding annotated image, where the colored bounding boxes indicate the annotated maintenance hole cover instances: green, good; red, broken; yellow, circle; blue, loose; purple, uncovered.

Model training was performed using the PyTorch deep learning framework on Ubuntu 22.04 (Canonical, London, UK), with a hardware configuration comprising an NVIDIA RTX 5060 Ti graphics card (NVIDIA, Santa Clara, CA, USA) and an Intel Core i7-14700 CPU (Intel, Santa Clara, CA, USA). All models were evaluated for parameters and floating-

point operations within a uniform hardware environment. The end-to-end inference latency and throughput of YOLO26-ABF, including preprocessing and postprocessing, were additionally measured on the same platform with a batch size of 1, and the results are reported in Table 2. The training parameters were established as follows: input image dimensions were 640 × 640; the AdamW optimizer was employed with an initial learning rate of 0.001 and a weight decay of 0.0005; the batch size was set to 16; and the total number of epochs was 300. All models were initialized from their official pretrained weights.

Table 2. End-to-end latency and throughput of YOLO26-ABF on the RTX 5060 Ti.

Platform	Precision	Export	Batch	End-to-End Latency (p50/p95)	FPS	Memory
RTX 5060 Ti	FP32	PyTorch	1	13 ms/44 ms	51	65 MB

The baseline detector is the nano-scale YOLO26 model (YOLO26-n) from the Ultralytics repository (ultralytics 8.4.22, PyTorch 2.10.0). Training was initialized from the official pretrained weights yolo26n.pt and the configuration file yolo26.yaml (YOLO26-n), with an input resolution of 640 × 640. The proposed YOLO26-ABF model was built on the modified configuration file yolo26-attn-bifpn-p2.yaml. All comparison models were trained under the identical protocol described above; no per-model hyperparameter search was performed. Table 3 summarizes the implementation details of the baseline model.

Table 3. Implementation details of the baseline model.

Item	Value
Model scale	YOLO26-n
Software	ultralytics 8.4.22 (PyTorch 2.10.0)
Source commit	1ce9d912e033fede606f02612ea1a364ce2e4d14 (ultralytics 8.4.22, local modified copy)
Pretrained weights	yolo26n.pt (Ultralytics official)
Configuration file	yolo26.yaml (YOLO26-n)
Input resolution	640 × 640

Table 4 summarizes the layer-level architecture of YOLO26-ABF. The network generates four feature levels (P2–P5). BiFPN performs weighted bidirectional fusion across these levels to refine the multiscale features, and the P2 level is incorporated as an additional high-resolution input for small-object detection. ASFF is applied before each of the three detection heads on the P3, P4, and P5 levels, and each head outputs both the class and bounding-box predictions. The three detection heads use a one-to-one assignment, and the inference is NMS-free.

Table 4. Layer-level architecture of YOLO26-ABF.

Feature Level	Input Size	Output Size	Channels	Operation	Detection Output
Backbone P2	640 × 640	160 × 160	64	Conv + C3k2	–
Backbone P3	160 × 160	80 × 80	128	Conv + C3k2	–
Backbone P4	80 × 80	40 × 40	128	Conv + C3k2	–
Backbone P5	40 × 40	20 × 20	256	Conv + C3k2 + SPPF + C2PSA	–
BiFPN fusion	P2–P5	P2–P5	128	weighted bidirectional fusion	–
Head P3	80 × 80	–	128	ASFF + one-to-one detection head (NMS-free)	class + box
Head P4	40 × 40	–	128	ASFF + one-to-one detection head (NMS-free)	class + box
Head P5	20 × 20	–	128	ASFF + one-to-one detection head (NMS-free)	class + box

3.2. Assessment Criteria

This work employed conventional quantitative criteria from the object detection domain to thoroughly test the YOLO26-ABF model's perceptual correctness and engineering potential across three dimensions: accuracy, complexity, and real-time performance.

Metrics of Accuracy: Precision, Recall, and Mean Average Precision (mAP). Among them, mAP_{50} assesses the model's classification accuracy at an IoU threshold of 0.5, whereas $mAP_{50–95}$ indicates the model's localization robustness over diverse IoU thresholds.

Complexity Metrics: Quantity of Parameters (Params) and Multiply–Accumulate Operations (GMACs). The quantity of parameters quantifies the model's memory overhead, represented in millions (M); GMACs quantify the overall computational demand per inference, indicating the model's hardware processing power requirements. All parameter counts and GMACs were measured with the thop library at an input resolution of 640×640.

For the real-time dimension, the end-to-end latency (including preprocessing, inference, and postprocessing) and memory consumption were measured on the RTX 5060 Ti with a batch size of 1. The average end-to-end latency was 19.5 ms (p50: 13 ms; p95: 44 ms), corresponding to a throughput of 51 FPS, and the peak GPU memory was 65 MB. The camera frame interval used as the real-time criterion is 33.3 ms, corresponding to an acquisition rate of 30 FPS. The p95 latency (44 ms) exceeds this frame interval; the real-time claim therefore refers to the average end-to-end latency.

Statistical Analysis: To account for training variability, key configurations were trained with multiple random seeds and the reported metrics are given as the mean $\pm$ standard deviation. Owing to the limited size of the fixed test set (168 images), 95% confidence intervals were estimated by bootstrapping the test images with 1000 resamples. For the final YOLO26-ABF model, the 95% confidence intervals were [0.90, 0.95] for mAP_{50} and [0.68, 0.75] for $mAP_{50–95}$. Performance differences were interpreted in terms of these intervals rather than point estimates alone. Because the comparative results in Sections 3.4 and 3.5 are based on single training runs, the corresponding differences are reported as point estimates and should be interpreted as indicative rather than as statistically confirmed improvements.

3.3. Ablation Experiments

This study performed ablation experiments using a 2×2 design to isolate the contributions of the architectural modification (BiFPN, the P2 high-resolution layer, and ASFF) and the data-driven Scheme D (Mosaic, MixUp, Copy–Paste, synthetic images, and pseudo-labeling). Four configurations were evaluated: the YOLO26 baseline, the baseline with the architectural modification only, the baseline with Scheme D only, and the complete YOLO26-ABF model. Each configuration was trained with multiple random seeds, and the mean and standard deviation are reported. The findings are presented in Table 5 and Figure 6.

The ablation results in Table 5 show that both the architectural modification and the data-driven Scheme D contribute to the final performance. Compared with the YOLO26 baseline, the architecture-only configuration (Exp. 2) improved mAP_{50} by 1 pp and $mAP_{50–95}$ by 3 pp, whereas the data-only configuration (Exp. 3) improved mAP_{50} by 1 pp and $mAP_{50–95}$ by 2 pp. The complete YOLO26-ABF model (Exp. 4) achieved mAP_{50} of 0.93 ± 0.007 and $mAP_{50–95}$ of 0.72 ± 0.010, corresponding to total gains of 3 pp and 6 pp over the baseline. These results indicate that the architectural and data-driven improvements are complementary: the architecture mainly enhances small-object feature fusion, while Scheme D primarily relieves class imbalance and enriches training diversity. The final YOLO26-ABF model's metrics were additionally assessed with bootstrap confidence intervals, as described in the Section 3.2.

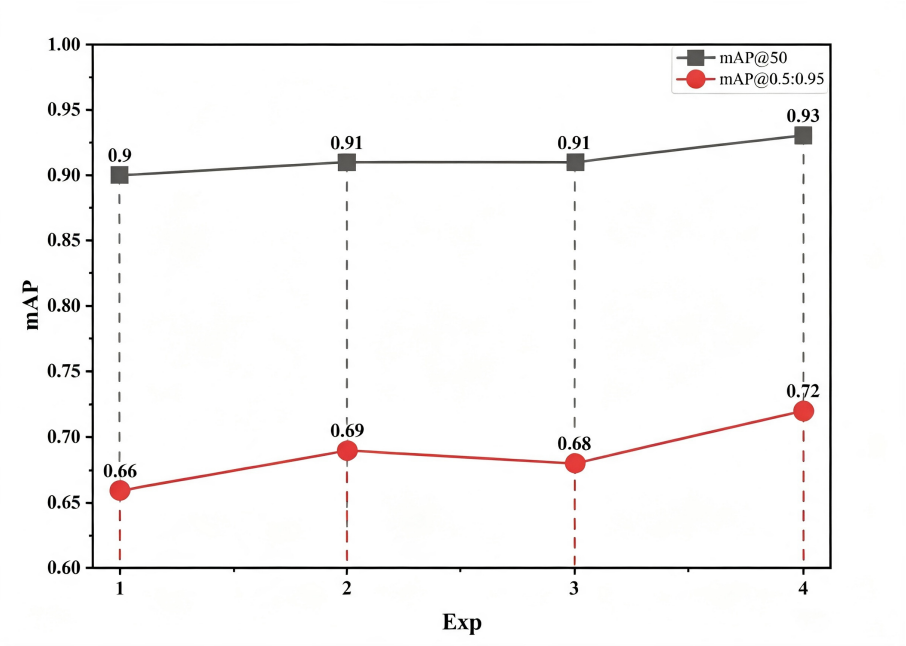


Figure 6. Ablation study of architecture and data contributions.

Table 5. Ablation study of architecture and data contributions (mean ± std over three seeds).

Exp.	YOLO26	Architecture	Scheme D	mAP_{50}	$mAP_{50–95}$
1	✓			0.90 ± 0.009	0.66 ± 0.012
2	✓	✓		0.91 ± 0.008	0.69 ± 0.011
3	✓		✓	0.91 ± 0.009	0.68 ± 0.013
4	✓	✓	✓	0.93 ± 0.007	0.72 ± 0.010

3.4. Comparative Analysis with Conventional Algorithms

This work conducted comparative trials using contemporary detection algorithms on a self-constructed dataset of maintenance hole covers to confirm the superiority of YOLO26-ABF. To assess the improvement over manhole-cover-specific detectors, we additionally retrained MCPH-YOLO [19] and an improved YOLOv8 detector [18] on the same dataset under the identical training protocol. The findings are presented in Table 6.

Table 6. Comparative analysis of overall performance across various object detection algorithms. Computational cost is reported in GMACs (1 GMAC = 2 GFLOPs).

Algorithm Model	Precision	Recall	mAP_{50}	$mAP_{50–95}$	Params/M	GMACs
YOLOv8	0.82	0.77	0.83	0.59	3.01	4.10
YOLOv11	0.87	0.83	0.87	0.63	2.60	3.25
MCPH-YOLO [19]	0.88	0.88	0.88	0.65	5.00	4.40
Improved YOLOv8 [18]	0.89	0.88	0.89	0.65	2.10	3.70
YOLO26	0.90	0.87	0.90	0.66	2.02	2.44
YOLO26-ABF	0.92	0.91	0.93	0.72	2.51	2.89

The comparative data analysis indicates that the traditional YOLOv8 model, constrained by its inadequate feature extraction from small objects, attained an mAP_{50} of merely 0.83. In contrast, YOLOv11 has undergone optimization of its feature network, improving its accuracy to 0.87; however, it continues to struggle in complex scenarios, such as detecting missing maintenance hole covers. The most recent benchmark, YOLO26, exhibits superior recognition capabilities, surpassing earlier models in accuracy and computational efficiency.

The YOLO26-ABF model presented in this study achieves an mAP_{50} of 0.93, which is 3 pp higher than the baseline YOLO26 in this comparison. Because the bootstrap 95% confidence interval for YOLO26-ABF is [0.90, 0.95] and the baseline point estimate is 0.90, this single-point difference overlaps the lower bound of the interval and is therefore indicative rather than a statistically confirmed gain. The improvement is accompanied by a moderate increase in the number of parameters (2.02 M to 2.51 M, +24.3%) and multiply–accumulate operations (2.44 to 2.89 GMACs, +18.4%), while the parameter count and computational cost of YOLO26-ABF remain lower than those of YOLOv8 and YOLOv11. With an average end-to-end latency of 19.5 ms (51 FPS) on the RTX 5060 Ti, the model satisfies the real-time criterion for 640 × 640 inputs at the camera acquisition rate of 30 FPS in terms of average end-to-end latency. Compared with MCPH-YOLO and the improved YOLOv8 detector, YOLO26-ABF improves mAP_{50} by 5.0 pp and 4.0 pp, respectively, in this single-run comparison; these differences are likewise reported as point estimates and should be interpreted with the same caution.

3.5. Analysis of Single-Class Detection Performance

This paper assessed the model's proficiency in fine-grained recognition of maintenance hole covers across various conditions—specifically, the identification of infrequent hazardous scenarios such as absent or misaligned covers—by calculating mAP_{50} for each of the five maintenance hole cover states and generating Precision–Recall curves. The findings are presented in Table 7 and Figure 7.

Table 7. Comparison of mAP_{50} detection outcomes by category.

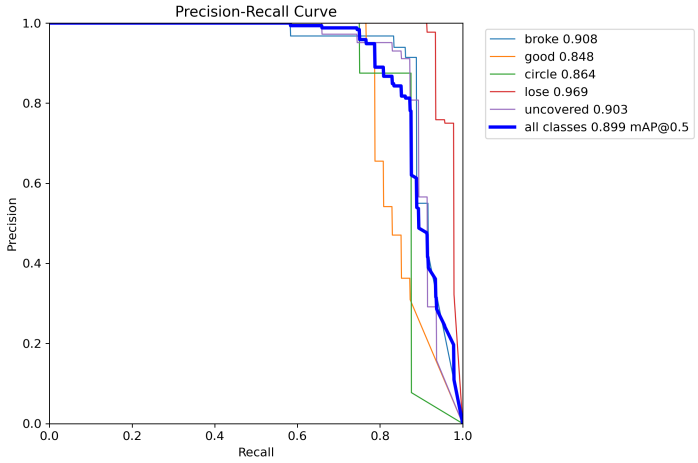
Class	YOLO26 mAP_{50}	YOLO26-ABF mAP_{50}	Improvement
good	0.85	0.90	+0.05
broken	0.91	0.93	+0.02
circle	0.86	0.89	+0.03
loose	0.97	0.97	0.00
uncovered	0.90	0.95	+0.05
Overall Mean	0.90	0.93	+0.03

Three principal implications can be derived from the single-class detection outcomes:

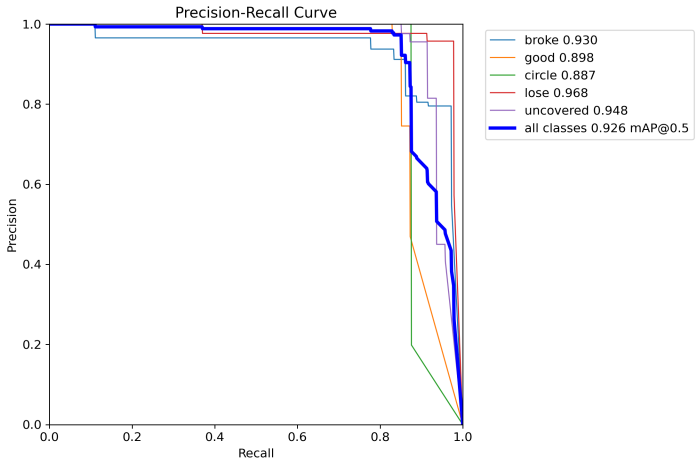
The identification of high-risk anomaly categories is highly reliable, and optimizing long-tail samples is crucial. In uncovered hazardous scenarios with a limited sample size, mAP_{50} rose from 0.90 to 0.95, reflecting a 5 pp difference in this single-run comparison, thereby confirming the efficacy of the synthetic image generation and semi-supervised pseudo-labeling methodologies presented in this paper. For the "loose" category, characterized by pronounced visual feature distinctiveness and a baseline accuracy of 0.97, the model's accuracy remained unchanged after the improvements. This illustrates the compatibility and robustness of the multi-module fusion methodology, encompassing BiFPN and ASFF. The final mAP_{50} for both hazardous-scenario categories exceeded 0.94, providing high-confidence perceptual support for the garbage truck's obstacle-avoidance decisions.

The model demonstrates exceptional overall detection stability. The Precision–Recall curves demonstrate that all categories are predominantly located in the upper-left corner of the coordinate system, indicating that the model achieves high precision and recall, with both false positives and false negatives kept at low levels. Notably, the mAP_{50} for the "loose" category attained 0.97—the highest across all categories—indicating that the model exhibits the most robust recognition capability for the "loose" scenario, which presents the greatest risk in sanitation operations, thereby aligning closely with the safety priority requirements of practical engineering applications.

The multiscale feature fusion module markedly enhances the detection of diminutive items. In the categories of "good," "broken," and "circle," which depend on edge textures and subtle deformations for differentiation, mAP_{50} increased by 5 pp, 2 pp, and 3 pp, respectively, in this single-run comparison, indicating a consistent pattern of improvement; these per-class differences are point estimates and were not separately tested for statistical significance. These findings confirm that the integrated architectural modification—combining BiFPN, the high-resolution P2 feature layer, and ASFF—effectively enhances the model's capacity to extract and differentiate subtle features, reduces feature loss in small maintenance hole covers at extended distances, and elevates the model's recognition accuracy for maintenance hole covers in various conditions.



(**a**)



(**b**)

Figure 7. Precision-Recall curves for the test dataset: (**a**) Baseline YOLO26 model; (**b**) YOLO26-ABF model. These curves were generated from a single training run (seed 0).

3.6. Analysis of Detection Performance by Object Size

To verify the effectiveness of the P2 high-resolution layer and the multiscale fusion modules for distant and small manhole covers, the test instances were divided into three groups of comparable size according to bounding-box area: Small (area $<$ 48,950 px^2; 61 instances), Medium (48,950–160,213 px^2; 61 instances), and Large (area $>$ 160,213 px^2; 62 instances). Table 8 reports the AP@50 of the baseline and YOLO26-ABF for each size group.

The results indicate that YOLO26-ABF achieves its largest gains on small objects, while the gains on medium and large objects are relatively smaller.

Table 8. Detection performance by object size on the test set.

Object Size	Instances	YOLO26 AP@50	YOLO26-ABF AP@50	Improvement
Small	61	0.82	0.87	+0.05
Medium	61	0.96	0.97	+0.01
Large	62	0.93	0.95	+0.02

3.7. Analysis of Model Error

This paper generated a normalized confusion matrix based on test-set findings (Figure 8) to uncover the model's identification deficiencies and misclassification patterns, and to assess the optimization effects of the improvement technique across hazard categories. The horizontal axis of the matrix denotes the actual condition of the maintenance hole cover, whilst the vertical axis indicates the model's predictive outcomes. The diagonal elements indicate the recall rates for the respective categories, whereas the off-diagonal elements denote the misclassification rates between categories.

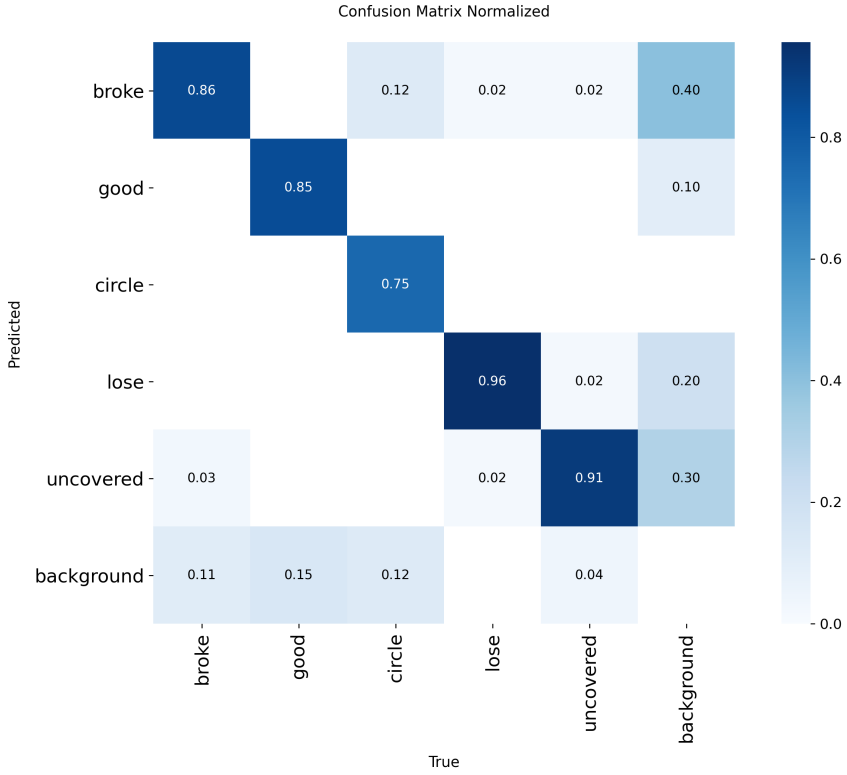


Figure 8. Normalized confusion matrix for the YOLO26-ABF model on the test dataset, generated from a single training run (seed 0).

Three principal conclusions can be derived from the results of the confusion matrix:

The model exhibits significant reliability in detecting high-risk anomaly categories. The recall rates for the two hazardous conditions—"loose" and "uncovered"—were 0.96 and 0.91, respectively, ranking among the highest across all categories. Additionally, the cross-class misclassification rate between these two categories was merely 2%, indicating minimal confusion with other states such as "good" or "broken." This outcome corroborates the findings from the Precision–Recall curve analysis and substantiates the efficacy of the synthetic image generation and semi-supervised pseudo-labeling strategies. Even with a limited sample size, the model successfully learns the essential features of hazardous categories, significantly mitigating the risk of overlooked or misclassified high-risk conditions and providing reliable perceptual support for the garbage truck's obstacle-avoidance decisions.

Secondly, there exists minor feature ambiguity among analogous states. The recall rate for "circle" is 0.75, the lowest among the five categories; the principal source of error is that 12% of samples were erroneously classified as "broken." This misclassification arises because some "circle" instances exhibit edge damage, leading to some overlap in visual features between the two categories and complicating classification. Overall, the rate of category confusion across states remains minimal, and no significant misclassification has occurred.

Third, the largest source of error arises from missed detections of small targets. A specific number of samples within each category were incorrectly identified as background (i.e., missed detections), with the "good" and "circle" categories demonstrating the highest false-negative rates at 15% and 12%, respectively. This error primarily arises from small, distant maintenance hole covers; the onboard camera's shooting angle and the vehicle's operating speed cause these covers to occupy a minimal portion of the image, leading to insufficient feature information and confusion with the road surface background.

The model's error distribution aligns with the characteristics of actual maintenance hole cover detection situations, and its elevated recognition accuracy in high-risk categories meets sanitation safety standards. The current challenges of failing to detect small targets and misidentifying similar states present critical opportunities for enhancement in forthcoming research.

3.8. Visual Examination of Detection Outcomes

To intuitively illustrate the efficacy of YOLO26-ABF in intricate operational scenarios, Figures 9–11 present the model's detection outcomes across various road surface conditions.

Figure 9 illustrates long-range detection of small maintenance hole covers by YOLO26-ABF.

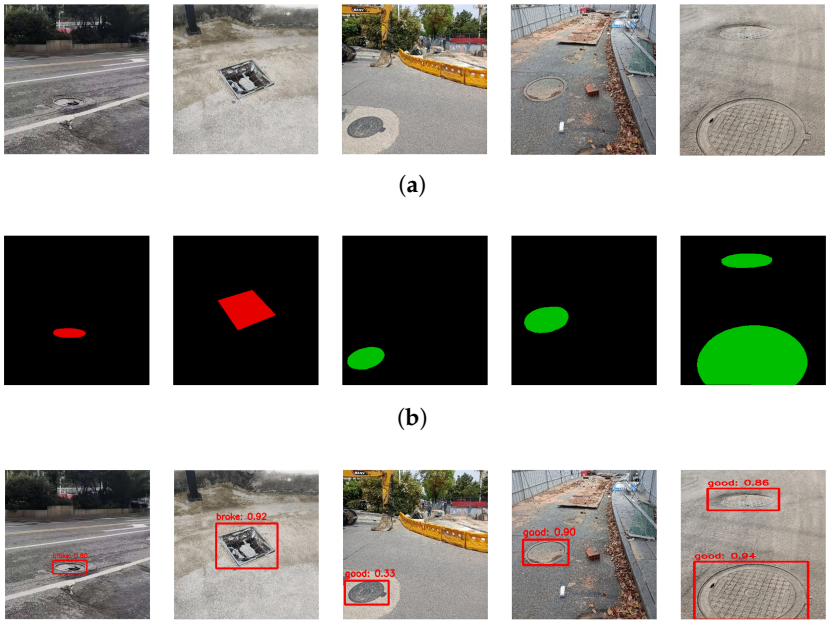


Figure 9. Long-range detection performance for small-target manhole covers: (**a**) original test images; (**b**) ground truth annotation results; (**c**) detection results of YOLO26-ABF model, where the colored bounding boxes indicate the maintenance hole cover states: green, good; red, broken; yellow, circle; blue, loose; purple, uncovered.

In perilous circumstances where maintenance hole covers are absent or exposed, the model achieves exceptional classification accuracy—significantly reducing the false positive rate—through sample augmentation with synthetic images and semi-supervised learning.

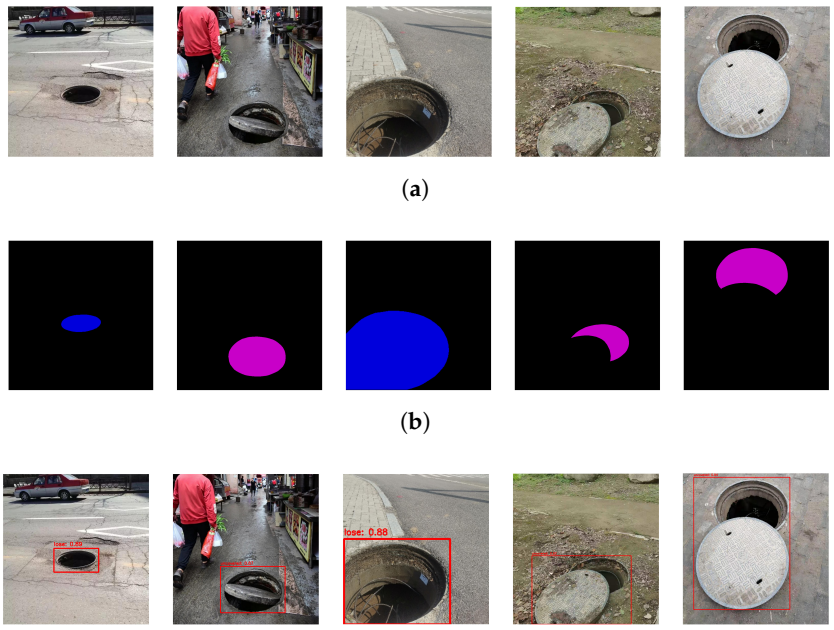


Figure 10. Illustration of maintenance hole cover detection under extremely dangerous situations: (**a**) original test images; (**b**) ground truth annotation results; (**c**) detection results of YOLO26-ABF model, where the colored bounding boxes indicate the maintenance hole cover states: green, good; red, broken; yellow, circle; blue, loose; purple, uncovered.

When the road surface is obscured by numerous fallen leaves, oil stains, or shadows, the integrated architectural modification allows the model to autonomously disregard extraneous background features, ensuring perceptual stability.

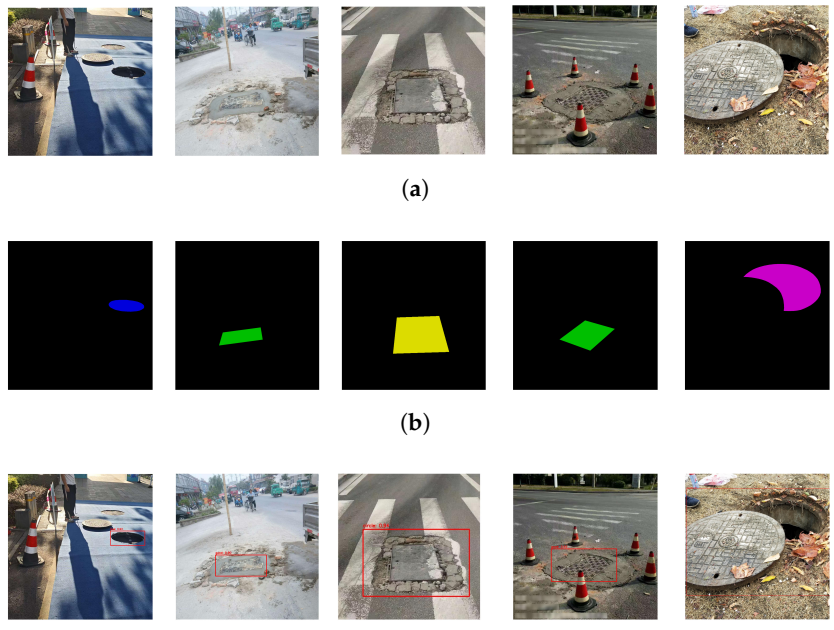


Figure 11. Maintenance hole cover detection outcomes in intricate backdrops: (**a**) original test images; (**b**) ground truth annotation results; (**c**) detection results of YOLO26-ABF model, where the colored bounding boxes indicate the maintenance hole cover states: green, good; red, broken; yellow, circle; blue, loose; purple, uncovered.

4. Discussion

This study develops and implements a real-time detection system using an enhanced YOLO26-ABF to address the challenge of detecting the maintenance hole cover state of garbage collection trucks during urban sanitation operations. The study successfully addresses industry challenges, including low recognition accuracy for small targets and a lack of outlier samples in complex contexts, by implementing comprehensive network architecture restructuring and multidimensional data augmentation. This paper's primary research conclusions are summarized as follows:

This study presents the ABF feature perception system, which addresses conflicts and dilution issues in multiscale feature propagation mathematically by incorporating the Weighted Bidirectional Feature Pyramid (BiFPN) and Adaptive Spatial Feature Fusion (ASFF) mechanisms. Simultaneously, incorporating the P2 high-resolution feature layer significantly enhances the model's ability to discern intricate edge details of diminutive maintenance hole covers at considerable distances. Empirical findings indicate that the YOLO26-ABF model achieves a mean Average Precision (mAP) of 0.93 on the proprietary dataset, demonstrating a substantial improvement over the baseline model.

The approach described in this paper achieves high accuracy with limited computational overhead, and the measured average end-to-end latency on the RTX 5060 Ti (19.5 ms, 51 FPS) is below the 33.3 ms frame interval of the 30 FPS camera, supporting real-time processing on average; the p95 latency (44 ms) exceeds the 33.3 ms frame interval, so the real-time claim is based on the mean latency. Although the model retains lightweight properties, deployment on embedded vehicle hardware has not yet been measured and remains future work. By accurately identifying five states of maintenance hole covers (good, broken, circle, loose, and uncovered), the method offers robust perceptual support for the vehicle's obstacle avoidance system, mitigating the risk of equipment damage or vehicle rollover incidents due to maintenance hole cover deficiencies, thereby ensuring the continuity and safety of urban sanitation operations.

The specific contribution of this work is the five-state manhole cover taxonomy and its evaluation from a refuse-collection-vehicle viewpoint, together with an integrated synthetic-image and semi-supervised data pipeline; the individual architectural components (BiFPN, P2, ASFF) are established techniques that we adapt and combine for this task.

To mitigate sample imbalance arising from extreme hazardous situations, such as missing or misplaced maintenance hole covers, this research employs an augmentation strategy that integrates synthetic image generation with semi-supervised pseudo-labeling methods. Synthetic image generation directly increases the sample size of critical categories, whereas pseudo-labeling mainly broadens the diversity of scenarios and road-surface materials. This strategy improves the model's adaptability to the diverse road-surface materials, occlusions, and lighting conditions encountered within the collected routes, thereby reducing the likelihood of missed detections and false positives during sanitation operations.

The YOLO26-ABF detection framework reliably recognizes the maintenance hole cover status of waste collection vehicles and provides an effective visual computing solution for automated oversight of smart city infrastructure. Future studies will investigate multi-sensor fusion perception technologies and refine the model for more severe extreme weather conditions to continuously improve the autonomous operational capabilities of garbage-collecting vehicles in all-weather environments.

Although the dataset was partitioned at the route level to avoid frame-level leakage, all images were acquired by one vehicle-mounted camera during a single collection campaign. The reported results therefore demonstrate performance on this campaign rather than cross-vehicle, cross-camera, or cross-region generalization. In addition, the safety-

related conclusions rest on a relatively narrow data base: 42 templates (22 "loose" and 20 "uncovered") were used to generate 515 synthetic training images, and the test set contains only 15 "loose" and 15 "uncovered" images; these limited sample sizes should be considered when interpreting the high-risk detection results. Independent evaluation on additional vehicles, cameras, routes, or publicly available manhole cover datasets is necessary to further validate the generalization of the proposed method and is planned as future work.

Author Contributions: Investigation, Z.C. (Zhongwen Chen) and Y.Z.; data curation, Z.C. (Zihua Chen); writing—original draft, Z.C. (Zihua Chen); writing—review and editing, Q.Z.; supervision, Q.Z. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The dataset was constructed in collaboration with a sanitation vehicle manufacturer and is part of the laboratory's proprietary resources; therefore, it cannot be made publicly available. Reasonable requests for collaborative research purposes can be directed to the corresponding author.

Acknowledgments: The authors thank the editor and reviewers for their valuable suggestions.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

ASFF	Adaptive Spatial Feature Fusion
BiFPN	Bidirectional Feature Pyramid Network
GFLOPs	Giga Floating-point Operations
GMACs	Giga Multiply–Accumulate Operations
IoT	Internet of Things
IoU	Intersection over Union
LiDAR	Light Detection and Ranging
NB-IoT	Narrowband Internet of Things
PAN	Path Aggregation Network
ReLU	Rectified Linear Unit

References

1. Li, J.; Qiao, Y.; Ma, H.; Mao, X. Design of City Manhole Cover Monitoring System Based on NB-IoT. In *Proceedings of the 2024 IEEE 7th Advanced Information Technology, Electronic and Automation Control Conference (IAEAC)*; IEEE: New York, NY, USA, 2024; Volume 7, pp. 870–874.
2. Guo, X.; Liu, B.; Wang, L. Design and implementation of intelligent manhole cover monitoring system based on NB-IoT. In *Proceedings of the 2019 International Conference on Robots & Intelligent System (ICRIS)*; IEEE: New York, NY, USA, 2019; pp. 207–210.
3. Mohanraj, S.; Aravindh, R.; Arshath Ahamed, N.; Arun, A. IoT based system for manhole monitoring and management. In *Proceedings of the 2023 9th International Conference on Advanced Computing and Communication Systems (ICACCS)*; IEEE: New York, NY, USA, 2023; Volume 1, pp. 1357–1361.
4. Donati, L.; Fontanini, T.; Tagliaferri, F.; Prati, A. An energy saving road sweeper using deep vision for garbage detection. *Appl. Sci.* **2020**, *10*, 8146. [CrossRef]
5. Liu, Y.; Ping, P.; Shi, Q.; Chen, H.; Yao, Q.; Luo, J. Research on the cleaning method of unmanned sweeper based on target distribution situation analysis. *Appl. Sci.* **2023**, *13*, 12544. [CrossRef]

6. Roche Cerasi, I. The potential of autonomous and connected sweepers for smart and sustainable cities. *Int. J. Transp. Dev. Integr.* **2022**, *6*, 37–57. [CrossRef]
7. Bele, K. Analysis of Impact of Maintenance Practices on Road Infrastructure. Ph.D. Thesis, University of Johannesburg, Johannesburg, South Africa, 2022.
8. Mo, F.; Yu, L.; Zhang, Z.; Zhao, Y. Design and implementation of manhole cover safety monitoring system based on smart light pole. *Math. Probl. Eng.* **2022**, *2022*, 3081649. [CrossRef]
9. Yu, Y.; Guan, H.; Ji, Z. Automated detection of urban road manhole covers using mobile laser scanning data. *IEEE Trans. Intell. Transp. Syst.* **2015**, *16*, 3258–3269. [CrossRef]
10. Wei, Z.; Yang, M.; Wang, L.; Ma, H.; Chen, X.; Zhong, R. Customized mobile LiDAR system for manhole cover detection and identification. *Sensors* **2019**, *19*, 2422. [CrossRef]
11. Ahmed, M.T.; Amin, M.R.; Rahman, F.U.; Abeer, M.H.; Nabi, N.; Ahamed, M.S. A Comparative Study of Different Transfer Learning Models for Classification of Urban Manhole States. In *Proceedings of the 2022 32nd International Conference on Computer Theory and Applications (ICCTA)*; IEEE: New York, NY, USA, 2022; pp. 25–30.
12. Hisban, M.A.M.; Norsahperi, N.M.H.; Khamis, N.; Elshishini, M.Y.; Ishak, A.J.; Ismail, L.I.; Hasan, W.Z.W. Enhancing Manhole Cover Detection and Location Management Using Modified YOLOv8 Algorithm: Performance Evaluation and Internet of Things Integration. In *Proceedings of the 2025 IEEE 16th Control and System Graduate Research Colloquium (ICSGRC)*; IEEE: New York, NY, USA, 2025; pp. 47–52.
13. Xu, Z.; Luo, H.; Zhu, H.; Sun, W.; Yang, S. YOLO-CSMD: Integrating Improved Convolutional Techniques for Manhole Cover Defect Detection. *IEEE Access* **2025**, *13*, 103155–103165. [CrossRef]
14. Xie, B.; Peng, K. Research on a Lightweight Manhole Cover Defect Detection Method Based on an Improved YOLOv5 Approach. In *Proceedings of the 2025 5th International Conference on High Performance Computing, Big Data and Communication Engineering (ICHBC)*; IEEE: New York, NY, USA, 2025; pp. 182–186.
15. Yang, L.; Hao, Z.; Hu, B.; Shan, C.; Wei, D.; He, D. Improved YOLOX-based detection of condition of road manhole covers. *Front. Built Environ.* **2024**, *10*, 1337984. [CrossRef]
16. Guo, D.; Xu, P.; Cai, M.; Liu, E.; Wang, M.; Shan, Z.; Jiang, F. DBG-YOLO: Efficient detection of hidden dangers of manhole covers based on deep learning YOLO network. *Multimed. Tools Appl.* **2025**, *84*, 46151–46172. [CrossRef]
17. Lin, X.; Yu, W.; Bai, J.; Cai, C. Improved Manhole Cover Defect Detection Using an Enhanced Backbone Architecture. In *Proceedings of the 2025 10th International Conference on Electronic Technology and Information Science (ICETIS)*; IEEE: New York, NY, USA, 2025; pp. 237–242.
18. Ding, Y.; Han, B.; Jiang, H.; Hu, H.; Xue, L.; Weng, J.; Tang, Z.; Liu, Y. Application of improved yolov8 image model in urban manhole cover defect management and detection: Case study. *Sensors* **2025**, *25*, 4144. [CrossRef]
19. Zeng, G.; Jiang, F.; Wang, M.; Zhang, S.; Wei, X.; Wu, G. Rapid detection and identification of road manhole cover safety hazards in complex scenarios. *Appl. Soft Comput.* **2025**, *182*, 113525. [CrossRef]
20. Pang, D.; Guan, Z.; Luo, T.; Su, W.; Dou, R. Real-time detection of road manhole covers with a deep learning model. *Sci. Rep.* **2023**, *13*, 16479. [CrossRef]
21. Mattheuwsen, L.; Vergauwen, M. Manhole cover detection on rasterized mobile mapping point cloud data using transfer learned fully convolutional neural networks. *Remote Sens.* **2020**, *12*, 3820. [CrossRef]
22. Liu, J.; Zhao, J.; Cao, Y.; Wang, Y.; Dong, C.; Guo, C. Road manhole cover defect detection via multi-scale edge enhancement and feature aggregation pyramid. *Sci. Rep.* **2025**, *15*, 10346. [CrossRef]
23. Zhang, D.; Yu, X.; Yang, L.; Quan, D.; Mi, H.; Yan, K. Data-augmented deep learning models for abnormal road manhole cover detection. *Sensors* **2023**, *23*, 2676. [CrossRef]
24. Hidayatullah, P.; Tubagus, R. YOLO26: A comprehensive architecture overview and key improvements. *arXiv* **2026**, arXiv:2602.14582.
25. Ultralytics. YOLO26. 2026. Available online: https://docs.ultralytics.com/models/yolo26/ (accessed on 15 July 2026).
26. Li, X.; Zhang, J.; Yang, Y.; Cheng, G.; Yang, K.; Tong, Y.; Tao, D. Sfnet: Faster and accurate semantic segmentation via semantic flow. *Int. J. Comput. Vis.* **2024**, *132*, 466–489. [CrossRef]
27. Tan, M.; Pang, R.; Le, Q.V. Efficientdet: Scalable and efficient object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*; IEEE: New York, NY, USA, 2020; pp. 10781–10790.
28. Zeng, S.; Yang, W.; Jiao, Y.; Geng, L.; Chen, X. Sca-YOLO: A New Small Object Detection Model for UAV Images. *Vis. Comput.* **2024**, *40*, 1787. [CrossRef]
29. Huang, X.; Wang, H.; Li, X. A multi-scale semantic feature fusion method for remote sensing crop classification. *Comput. Electron. Agric.* **2024**, *224*, 109185. [CrossRef]

30. Jin, X.; Zhu, H.; Li, S.; Wang, Z.; Liu, Z.; Tian, J.; Yu, C.; Qin, H.; Li, S.Z. A survey on mixup augmentations and beyond. *arXiv* **2024**, arXiv:2409.05202.
31. Liu, Y.; Ma, C.; He, Z.; Kuo, C.; Chen, K.; Zhang, P.; Wu, B.; Kira, Z.; Vajda, P. Unbiased teacher for semi-supervised object detection. *arXiv* **2021**, arXiv:2102.09480.

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